

Exploratory Data Analysis

Understanding the dataset to explore how the data is present in the database and if there is a need of creating some aggregated tables that can help with:

- Vendor selection for profitability
- Product Pricing Optimization

```
In [2]: import pandas as pd
import sqlite3
```

```
In [3]: conn = sqlite3.connect("inventory.db")
```

```
In [4]: #To check which tables are present in your SQLite database
tables = pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)
tables
```

```
Out[4]:
```

	name
0	begin_inventory-checkpoint
1	begin_inventory
2	end_inventory
3	purchases
4	purchase_prices
5	sales
6	vendor_invoice

```
In [5]: # Loop through all tables
for table in tables['name']:

    print("\n" + "="*50)
    print(f"Table Name: {table}")
    print("="*50)

    try:
        # Fetch first 5 rows
        df = pd.read_sql(
            f'SELECT * FROM "{table}" LIMIT 5;',
            conn
        )

        # Display dataframe
        display(df)

    except Exception as e:
```

```
print(f"Error reading {table}: {e}")
```

```
=====
Table Name: begin_inventory-checkpoint
=====
```

	InventoryId	Store	City	Brand	Description	Size	onHand	Price	s
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	750mL	8	12.99	20;
1	1_HARDERSFIELD_60	1	HARDERSFIELD	60	Canadian Club 1858 VAP	750mL	7	10.99	20;
2	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	750mL	6	36.99	20;
3	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	750mL	3	38.99	20;
4	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	750mL	6	34.99	20;

```
=====
Table Name: begin_inventory
=====
```

	InventoryId	Store	City	Brand	Description	Size	onHand	Price	s
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	750mL	8	12.99	20;
1	1_HARDERSFIELD_60	1	HARDERSFIELD	60	Canadian Club 1858 VAP	750mL	7	10.99	20;
2	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	750mL	6	36.99	20;
3	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	750mL	3	38.99	20;
4	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	750mL	6	34.99	20;

```
=====
Table Name: end_inventory
=====
```

	InventoryId	Store	City	Brand	Description	Size	onHand	Price	
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	750mL	11	12.99	20%
1	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	750mL	7	36.99	20%
2	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	750mL	7	38.99	20%
3	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	750mL	4	34.99	20%
4	1_HARDERSFIELD_75	1	HARDERSFIELD	75	Three Olives Tomato Vodka	750mL	7	14.99	20%

=====

Table Name: purchases

=====

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorName
0	69_MOUNTMEND_8412	69	8412	Tequila Ocho Plata Fresno	750mL	105	ALTAMAR BRANDS LLC
1	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE
2	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE
3	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE
4	76_DONCASTER_2034	76	2034	Glendalough Double Barrel	750mL	388	ATLANTIC IMPORTING COMPANY

=====

Table Name: purchase_prices

=====

	Brand	Description	Price	Size	Volume	Classification	PurchasePrice	VendorNumber
0	58	Gekkeikan Black & Gold Sake	12.99	750mL	750	1	9.28	8320
1	62	Herradura Silver Tequila	36.99	750mL	750	1	28.67	1128
2	63	Herradura Reposado Tequila	38.99	750mL	750	1	30.46	1128
3	72	No. 3 London Dry Gin	34.99	750mL	750	1	26.11	9165
4	75	Three Olives Tomato Vodka	14.99	750mL	750	1	10.94	7245

=====
Table Name: sales
=====

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	Sale
0	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
1	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	2	32.98	
2	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
3	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	14.49	
4	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	2	69.98	

=====
Table Name: vendor_invoice
=====

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
0	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023-12-21	2024-02-16	6
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	15
2	388	ATLANTIC IMPORTING COMPANY	2024-01-09	8169	2023-12-24	2024-02-16	5
3	480	BACARDI USA INC	2024-01-12	8106	2023-12-20	2024-02-05	10100
4	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023-12-24	2024-02-12	1935

```
In [6]: purchases = pd.read_sql(  
        'SELECT * FROM "purchases" WHERE VendorNumber = 4466;',  
        conn  
    )  
purchases
```

Out[6]:

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorNam
0	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICA VINTAG BEVERAG
1	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
2	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICA VINTAG BEVERAG
3	38_GOULCREST_5215	38	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
4	59_CLAETHORPES_5215	59	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
...	...	...	...	...	...	...	
2187	81_PEMBROKE_5215	81	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
2188	62_KILMARNOCK_5255	62	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICA VINTAG BEVERAG
2189	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
2190	6_GOULCREST_5215	6	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICA VINTAG BEVERAG
2191	35_HALIVAARA_5255	35	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICA VINTAG BEVERAG

2192 rows × 16 columns

```
In [32]: purchase_price = pd.read_sql( 'SELECT * FROM "purchase_prices" WHERE VendorNumber  
purchase_price
```

Out[32]:

	Brand	Description	Price	Size	Volume	Classification	PurchasePrice	VendorNumbe
0	5215	TGI Fridays Long Island Iced	12.99	1750mL	1750	1	9.41	446
1	5255	TGI Fridays Ultimte Mudslide	12.99	1750mL	1750	1	9.35	446
2	3140	TGI Fridays Orange Dream	14.99	1750mL	1750	1	11.19	446

```
In [17]: vendor_invoice = pd.read_sql(  
          'SELECT * FROM "vendor_invoice" WHERE VendorNumber = 4466;',  
          conn  
        )  
  
        vendor_invoice
```

Out[17]:	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
0	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-19	8207	2023-12-27	2024-02-26	3
2	4466	AMERICAN VINTAGE BEVERAGE	2024-01-18	8307	2024-01-03	2024-02-18	
3	4466	AMERICAN VINTAGE BEVERAGE	2024-01-27	8469	2024-01-14	2024-03-11	
4	4466	AMERICAN VINTAGE BEVERAGE	2024-02-04	8532	2024-01-19	2024-03-15	
5	4466	AMERICAN VINTAGE BEVERAGE	2024-02-09	8604	2024-01-24	2024-03-15	3
6	4466	AMERICAN VINTAGE BEVERAGE	2024-02-17	8793	2024-02-05	2024-04-02	
7	4466	AMERICAN VINTAGE BEVERAGE	2024-03-01	8892	2024-02-12	2024-03-28	1
8	4466	AMERICAN VINTAGE BEVERAGE	2024-03-07	8995	2024-02-19	2024-04-02	1
9	4466	AMERICAN VINTAGE BEVERAGE	2024-03-12	9033	2024-02-22	2024-04-16	1
10	4466	AMERICAN VINTAGE BEVERAGE	2024-03-16	9180	2024-03-03	2024-04-19	2
11	4466	AMERICAN VINTAGE BEVERAGE	2024-03-23	9244	2024-03-08	2024-04-21	1
12	4466	AMERICAN VINTAGE BEVERAGE	2024-03-31	9371	2024-03-17	2024-05-13	1
13	4466	AMERICAN VINTAGE BEVERAGE	2024-04-09	9491	2024-03-24	2024-05-08	2

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
14	4466	AMERICAN VINTAGE BEVERAGE	2024-04-17	9583	2024-03-31	2024-05-12	1
15	4466	AMERICAN VINTAGE BEVERAGE	2024-04-20	9639	2024-04-04	2024-06-04	5
16	4466	AMERICAN VINTAGE BEVERAGE	2024-04-29	9800	2024-04-15	2024-06-07	2
17	4466	AMERICAN VINTAGE BEVERAGE	2024-05-09	9886	2024-04-21	2024-06-12	3
18	4466	AMERICAN VINTAGE BEVERAGE	2024-05-14	9999	2024-04-29	2024-06-26	3
19	4466	AMERICAN VINTAGE BEVERAGE	2024-05-16	10095	2024-05-06	2024-06-27	2
20	4466	AMERICAN VINTAGE BEVERAGE	2024-05-28	10169	2024-05-11	2024-07-04	3
21	4466	AMERICAN VINTAGE BEVERAGE	2024-06-04	10257	2024-05-17	2024-07-08	3
22	4466	AMERICAN VINTAGE BEVERAGE	2024-06-12	10346	2024-05-23	2024-07-20	6
23	4466	AMERICAN VINTAGE BEVERAGE	2024-06-17	10445	2024-05-30	2024-07-19	2
24	4466	AMERICAN VINTAGE BEVERAGE	2024-06-22	10600	2024-06-09	2024-08-01	3
25	4466	AMERICAN VINTAGE BEVERAGE	2024-06-29	10695	2024-06-16	2024-08-12	1
26	4466	AMERICAN VINTAGE BEVERAGE	2024-07-09	10777	2024-06-22	2024-08-15	
27	4466	AMERICAN VINTAGE BEVERAGE	2024-07-08	10836	2024-06-25	2024-08-14	

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
28	4466	AMERICAN VINTAGE BEVERAGE	2024-07-11	10969	2024-06-29	2024-08-18	
29	4466	AMERICAN VINTAGE BEVERAGE	2024-07-19	11085	2024-07-06	2024-09-04	6
30	4466	AMERICAN VINTAGE BEVERAGE	2024-07-26	11187	2024-07-14	2024-08-26	15
31	4466	AMERICAN VINTAGE BEVERAGE	2024-08-03	11244	2024-07-18	2024-09-04	2
32	4466	AMERICAN VINTAGE BEVERAGE	2024-08-11	11362	2024-07-26	2024-09-24	2
33	4466	AMERICAN VINTAGE BEVERAGE	2024-08-18	11489	2024-08-04	2024-09-16	7
34	4466	AMERICAN VINTAGE BEVERAGE	2024-08-24	11540	2024-08-08	2024-10-02	12
35	4466	AMERICAN VINTAGE BEVERAGE	2024-09-02	11716	2024-08-19	2024-09-29	4
36	4466	AMERICAN VINTAGE BEVERAGE	2024-09-12	11771	2024-08-23	2024-10-11	3
37	4466	AMERICAN VINTAGE BEVERAGE	2024-09-20	11901	2024-09-01	2024-10-30	3
38	4466	AMERICAN VINTAGE BEVERAGE	2024-09-25	11993	2024-09-07	2024-10-23	2
39	4466	AMERICAN VINTAGE BEVERAGE	2024-10-01	12125	2024-09-16	2024-11-07	2
40	4466	AMERICAN VINTAGE BEVERAGE	2024-10-08	12235	2024-09-23	2024-11-20	2
41	4466	AMERICAN VINTAGE BEVERAGE	2024-10-09	12253	2024-09-23	2024-11-14	

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
42	4466	AMERICAN VINTAGE BEVERAGE	2024-10-12	12321	2024-09-26	2024-11-19	1
43	4466	AMERICAN VINTAGE BEVERAGE	2024-10-20	12466	2024-10-05	2024-11-26	2
44	4466	AMERICAN VINTAGE BEVERAGE	2024-10-27	12515	2024-10-09	2024-11-30	1
45	4466	AMERICAN VINTAGE BEVERAGE	2024-11-07	12702	2024-10-21	2024-12-11	1
46	4466	AMERICAN VINTAGE BEVERAGE	2024-11-12	12752	2024-10-25	2024-12-11	2
47	4466	AMERICAN VINTAGE BEVERAGE	2024-11-20	12828	2024-10-30	2024-12-18	2
48	4466	AMERICAN VINTAGE BEVERAGE	2024-11-27	12929	2024-11-06	2025-01-04	2
49	4466	AMERICAN VINTAGE BEVERAGE	2024-11-28	13092	2024-11-16	2024-12-30	2
50	4466	AMERICAN VINTAGE BEVERAGE	2024-12-06	13134	2024-11-20	2025-01-18	3
51	4466	AMERICAN VINTAGE BEVERAGE	2024-12-16	13254	2024-11-28	2025-01-13	2
52	4466	AMERICAN VINTAGE BEVERAGE	2024-12-26	13432	2024-12-09	2025-01-27	2
53	4466	AMERICAN VINTAGE BEVERAGE	2024-12-30	13483	2024-12-13	2025-02-11	2
54	4466	AMERICAN VINTAGE BEVERAGE	2025-01-09	13627	2024-12-22	2025-02-05	4

```
In [8]: purchases.groupby(['Brand', 'PurchasePrice'])[['Quantity', 'Dollars']].sum()
```

Out[8]:

		Quantity	Dollars
Brand	PurchasePrice		

Brand	PurchasePrice		
3140	11.19	4640	51921.60
5215	9.41	4923	46325.43
5255	9.35	6215	58110.25

In [9]: `vendor_invoice['PONumber'].nunique()`

Out[9]: 55

In [10]: `vendor_invoice['PONumber'].shape`

Out[10]: (55,)

```
In [17]: sales_summary = pd.read_sql(
    '''
    SELECT
        Brand,
        SUM(SalesDollars) AS SalesDollars,
        SUM(SalesPrice) AS SalesPrice,
        SUM(SalesQuantity) AS SalesQuantity
    FROM sales
    GROUP BY Brand
    ''',
    conn
)

sales_summary.head()
```

Out[17]:

	Brand	SalesDollars	SalesPrice	SalesQuantity
--	-------	--------------	------------	---------------

0	58	43341.54	28145.64	3446
1	60	18716.25	10720.79	1775
2	61	4364.88	363.74	312
3	62	119863.75	90154.51	3125
4	63	112249.22	88553.10	2778

- The `purchases` table contains actual purchase data, including the date of purchase, products (brands) purchased by vendors, the amount paid (in dollars), and the quantity purchased.
- The purchase price column is derived from the `purchase_prices` table, which provides product-wise actual and purchase prices. The combination of vendor and brand is unique in this table.

- The `vendor_invoice` table aggregates data from the `purchases` table, summarizing quantity and dollar amounts, along with an additional column for freight. This table maintains uniqueness based on vendor and PO number.
 - The `sales` table captures actual sales transactions, detailing the brands purchased by vendors, the quantity sold, the selling price, and the revenue earned.
-

As the data required for analysis is distributed across different tables, we need to create a summary table containing:

- purchase transactions made by vendors
- sales transaction data
- freight costs for each vendor
- actual product prices from vendors

```
In [18]: vendor_invoice.columns
```

```
Out[18]: Index(['VendorNumber', 'VendorName', 'InvoiceDate', 'PONumber', 'PODate',  
              'PayDate', 'Quantity', 'Dollars', 'Freight', 'Approval'],  
              dtype='object')
```

```
In [19]: freight_summary = pd.read_sql_query("""select VendorNumber, SUM(Freight) as Freight  
FROM vendor_invoice  
GROUP BY VendorNumber""", conn)
```

```
In [20]: freight_summary
```

```
Out[20]:
```

	VendorNumber	FreightCost
0	2	27.08
1	54	0.48
2	60	367.52
3	105	62.39
4	200	6.19
...	...	...
121	98450	856.02
122	99166	130.09
123	172662	178.34
124	173357	202.50
125	201359	0.09

126 rows × 2 columns

```
In [34]: print(purchases.columns.tolist())
print(purchase_price.columns.tolist())
```

```
['InventoryId', 'Store', 'Brand', 'Description', 'Size', 'VendorNumber', 'VendorName',
 'PONumber', 'PODate', 'ReceivingDate', 'InvoiceDate', 'PayDate', 'PurchasePrice',
 'Quantity', 'Dollars', 'Classification']
['Brand', 'Description', 'Price', 'Size', 'Volume', 'Classification', 'PurchasePrice',
 'VendorNumber', 'VendorName']
```

```
In [ ]: pd.read_sql_query("""
SELECT
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.PurchasePrice,
    pp.Volume,
    pp.Price AS ActualPrice,
    SUM(p.Quantity) AS TotalPurchaseQuantity,
    SUM(p.Dollars) AS TotalPurchaseDollars

FROM purchases p

JOIN purchase_prices pp
    ON p.Brand = pp.Brand

WHERE p.PurchasePrice > 0

GROUP BY
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.PurchasePrice,
    pp.Volume,
    pp.Price

ORDER BY TotalPurchaseDollars
""", conn)
```

```
In [4]: pd.read_sql_query("""
SELECT
    VendorNo,
    Brand,
    SUM(ExciseTax) AS TotalExciseTax,
    SUM(SalesDollars) AS TotalSalesDollars,
    SUM(SalesPrice) AS TotalSalesPrice,
    SUM(SalesQuantity) AS TotalSalesQuantity
FROM sales
GROUP BY VendorNo, Brand
""", conn)
```

Out[4]:

	VendorNo	Brand	TotalExciseTax	TotalSalesDollars	TotalSalesPrice	TotalSalesQuant
0	2	90085	2.00	665.82	295.92	
1	2	90609	0.52	599.76	449.82	
2	60	771	37.01	704.53	494.67	
3	60	3979	7224.06	66871.69	41682.51	39
4	105	2529	9.44	359.88	59.98	
...	...	...	...	...	...	
11267	173357	2804	110.33	6298.60	3194.29	1
11268	173357	3666	141.19	8996.40	4873.05	1
11269	173357	3848	4.71	185.94	92.97	
11270	173357	3909	773.87	24540.18	14469.21	9
11271	201359	90609	1.24	1474.41	1324.47	

11272 rows × 6 columns

In [5]:

```
pd.read_sql_query(  
    'PRAGMA table_info("sales")',  
    conn  
)
```

```
Out[5]:
```

	cid	name	type	notnull	dflt_value	pk
0	0	InventoryId	TEXT	0	None	0
1	1	Store	BIGINT	0	None	0
2	2	Brand	BIGINT	0	None	0
3	3	Description	TEXT	0	None	0
4	4	Size	TEXT	0	None	0
5	5	SalesQuantity	BIGINT	0	None	0
6	6	SalesDollars	FLOAT	0	None	0
7	7	SalesPrice	FLOAT	0	None	0
8	8	SalesDate	TEXT	0	None	0
9	9	Volume	FLOAT	0	None	0
10	10	Classification	BIGINT	0	None	0
11	11	ExciseTax	FLOAT	0	None	0
12	12	VendorNo	BIGINT	0	None	0
13	13	VendorName	TEXT	0	None	0

```
In [6]: pd.read_sql_query(  
        "SELECT COUNT(*) AS TotalRows FROM sales",  
        conn  
    )
```

```
Out[6]:
```

	TotalRows
0	12825363

```
In [ ]:
```

```
In [7]: vendor_sales_summary = pd.read_sql_query("""  
        WITH FreightSummary AS (  
            SELECT  
                VendorNumber,  
                SUM(Freight) AS FreightCost  
            FROM vendor_invoice  
            GROUP BY VendorNumber  
        ),  
        PurchaseSummary AS (  
            SELECT  
                p.VendorNumber,  
                p.VendorName,  
                p.Brand,  
                p.Description,
```



```
        pp.Volume,
        pp.PurchasePrice,
        pp.Price AS ActualPrice,
        SUM(p.Quantity) AS TotalPurchaseQuantity,
        SUM(p.Dollars) AS TotalPurchaseDollars
FROM purchases p
JOIN purchase_prices pp
    ON p.Brand = pp.Brand
GROUP BY
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.Description,
    pp.Volume,
    pp.PurchasePrice,
    pp.Price
),

SalesSummary AS (
    SELECT
        VendorNo,
        Brand,
        SUM(SalesQuantity) AS TotalSalesQuantity,
        SUM(SalesDollars) AS TotalSalesDollars,
        SUM(SalesPrice) AS TotalSalesPrice,
        SUM(ExciseTax) AS TotalExciseTax
    FROM sales
    GROUP BY VendorNo, Brand
)

SELECT
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.Description,
    p.Volume,
    p.PurchasePrice,
    p.ActualPrice,

    p.TotalPurchaseQuantity,
    p.TotalPurchaseDollars,

    s.TotalSalesQuantity,
    s.TotalSalesDollars,
    s.TotalSalesPrice,
    s.TotalExciseTax,

    f.FreightCost

FROM PurchaseSummary p

LEFT JOIN SalesSummary s
    ON p.VendorNumber = s.VendorNo
    AND p.Brand = s.Brand

LEFT JOIN FreightSummary f
```

```
ON p.VendorNumber = f.VendorNumber

ORDER BY p.TotalPurchaseDollars DESC
""", conn)
```

In [8]: vendor_sales_summary

Out[8]:

	VendorNumber	VendorName	Brand	Description	Volume	PurchasePrice	Actual
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	1750	26.27	
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	1750	23.19	
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	1750	18.24	
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	1750	16.17	
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	1750	21.89	
...	...	...	...	...	...	...	...
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	50	0.39	
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	50	0.74	
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	200	1.47	
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	50	0.71	
10692	2561	EDRINGTON AMERICAS	2166	The Macallan Double Cask 12	750	0.00	

10693 rows × 14 columns

In [9]: vendor_sales_summary.dtypes

```
Out[9]: VendorNumber      int64
VendorName      object
Brand          int64
Description     object
Volume         object
PurchasePrice   float64
ActualPrice     float64
TotalPurchaseQuantity  int64
TotalPurchaseDollars  float64
TotalSalesQuantity  float64
TotalSalesDollars    float64
TotalSalesPrice     float64
TotalExciseTax       float64
FreightCost         float64
dtype: object
```

```
In [12]: vendor_sales_summary.isnull().sum()
```

```
Out[12]: VendorNumber      0
VendorName      0
Brand          0
Description     0
Volume         0
PurchasePrice   0
ActualPrice     0
TotalPurchaseQuantity  0
TotalPurchaseDollars  0
TotalSalesQuantity  0
TotalSalesDollars    0
TotalSalesPrice     0
TotalExciseTax       0
FreightCost         0
dtype: int64
```

```
In [11]: vendor_sales_summary.fillna(0, inplace=True)
```

```
In [13]: vendor_sales_summary['VendorName'].unique()
```

```

Out[13]: array(['BROWN-FORMAN CORP', 'MARTIGNETTI COMPANIES',
                'PERNOD RICARD USA', 'DIAGEO NORTH AMERICA INC',
                'BACARDI USA INC', 'JIM BEAM BRANDS COMPANY',
                'MAJESTIC FINE WINES', 'ULTRA BEVERAGE COMPANY LLP',
                'STOLI GROUP,(USA) LLC', 'PROXIMO SPIRITS INC.',
                'MOET HENNESSY USA INC', 'CAMPARI AMERICA',
                'SAZERAC CO INC', 'CONSTELLATION BRANDS INC',
                'M S WALKER INC', 'SAZERAC NORTH AMERICA INC.',
                'PALM BAY INTERNATIONAL INC', 'REMY COINTREAU USA INC',
                'SIDNEY FRANK IMPORTING CO', 'E & J GALLO WINERY',
                'WILLIAM GRANT & SONS INC', 'HEAVEN HILL DISTILLERIES',
                'DISARONNO INTERNATIONAL LLC', 'EDRINGTON AMERICAS',
                'CASTLE BRANDS CORP.', 'SOUTHERN WINE & SPIRITS NE',
                'STE MICHELLE WINE ESTATES', 'TRINCHERO FAMILY ESTATES',
                'MHW LTD', 'WINE GROUP INC',
                'PERFECTA WINES', 'LUXCO INC',
                'TREASURY WINE ESTATES', 'DIAGEO CHATEAU ESTATE WINES',
                'SHAW ROSS INT L IMP LTD', 'PINE STATE TRADING CO',
                'PHILLIPS PRODUCTS CO.', 'CALEDONIA SPIRITS INC',
                'STATE WINE & SPIRITS', 'KOBRA BRAND CORPORATION',
                'BANFI PRODUCTS CORP', 'VINEYARD BRANDS INC',
                'DELICATO VINEYARDS INC', 'FABRIZIA SPIRITS LLC',
                'DUGGANS DISTILLED PRODUCTS', 'Serralles Usa LLC',
                'SEA HAGG DISTILLERY LLC', 'OLE SMOKY DISTILLERY LLC',
                'VRANKEN AMERICA', 'KLIN SPIRITS LLC',
                'LAIRD & CO', 'ADAMBA IMPORTS INTL INC',
                'LATITUDE BEVERAGE COMPANY', 'FREDERICK WILDMAN & SONS',
                'MCCORMICK DISTILLING CO', 'CHARLES JACQUIN ET CIE INC',
                'WESTERN SPIRITS BEVERAGE CO', 'MARSALLE COMPANY',
                'AMERICAN VINTAGE BEVERAGE', 'MANGO BOTTLING INC',
                'SWEET BABY VINEYARD', 'NICHE W & S',
                'LABELLE VYDS AND WINERY', 'FLAG HILL WINERY & VINEYARD',
                'SMOKY QUARTZ DISTILLERY LLC', 'PREMIUM PORT WINES, INC.',
                'Russian Standard Vodka', 'Dunn Wine Brokers',
                'WEIN BAUER INC', 'BULLY BOY DISTILLERS',
                'ATLANTIC IMPORTING COMPANY', 'PREMIER DISTRIBUTORS',
                'VINILANDIA USA', 'PARK STREET IMPORTS LLC',
                'TAKARA SAKE USA INC', 'SEA BREEZE CELLARS LLC',
                'STARK BREWING COMPANY', 'TY KU LLC',
                'PSP WINES', 'TAMWORTH DISTILLING',
                'ZORVINO VINEYARDS', 'SOUTHERN GLAZERS W&S OF NE',
                'HOOD RIVER DISTILLERS, Inc.', 'CRUSH WINES',
                'POVERTY LANE ORCHARDS', 'DJINN SPIRITS LLC',
                'MOONLIGHT MEADERY', 'TALL SHIP DISTILLERY LLC',
                'FORTUNE WINE BROKERS LLC', 'BLACK COVE BEVERAGES',
                'VINEXTRA INC', 'SURVILLE ENTERPRISES CORP',
                'JEWELL TOWNE VINEYARDS', 'SWEETWATER FARM',
                'MARTIGNETTI COMPANIES', 'ALTAMAR BRANDS LLC',
                'CANDIA VINEYARDS', 'INCREDIBREW INC',
                'ALISA CARR BEVERAGES', 'STELLAR IMPORTING CO LLC',
                'FULCHINO VINEYARD INC',
                'IRA GOLDMAN AND WILLIAMS, LLP',
                'Circa Wines', 'VINEDREA WINES LLC',
                'BLACK PRINCE DISTILLERY INC', 'VINEYARD BRANDS LLC',
                'THE IMPORTED GRAPE LLC', 'WALPOLE MTN VIEW WINERY',
                'GILMANTON WINERY & VINEYARD', 'HAUNTING WHISPER VYDS',

```

```

        'STAR INDUSTRIES INC.', 'LOYAL DOG WINERY',
        'R.P.IMPORTS INC', 'THE PIERPONT GROUP LLC',
        'APPOLO VINEYARDS LLC', 'BLACK ROCK SPIRITS LLC',
        'CENTEUR IMPORTS LLC', 'HIGHLAND WINE MERCHANTS LLC',
        'AMERICAN SPIRITS EXCHANGE', 'UNCORKED',
        'BRONCO WINE COMPANY', 'MILTONS DISTRIBUTING CO',
        'TRUETT HURST', 'LAUREATE IMPORTS CO',
        'FANTASY FINE WINES CORP', 'AAPER ALCOHOL & CHEMICAL CO',
        'SILVER MOUNTAIN CIDERS', 'CAPSTONE INTERNATIONAL',
        'FLAVOR ESSENCE INC', ], dtype=object)

```

```
In [14]: vendor_sales_summary['Description'].unique()
```

```
Out[14]: array(['Jack Daniels No 7 Black', "Tito's Handmade Vodka",
        'Absolut 80 Proof', ..., 'Concannon Glen Ellen Wh Zin',
        'The Club Strawbry Margarita', 'The Macallan Double Cask 12'],
        shape=(9652,), dtype=object)
```

```
In [15]: vendor_sales_summary['Volume'] = vendor_sales_summary['Volume'].astype('float64')
```

```
In [16]: vendor_sales_summary['VendorName'] = vendor_sales_summary['VendorName'].str.strip()
```

```
In [17]: vendor_sales_summary['GrossProfit'] = vendor_sales_summary['TotalSalesDollars'] -
```

```
In [18]: vendor_sales_summary['GrossProfit'].min()
```

```
Out[18]: -52002.780000000006
```

```
In [19]: vendor_sales_summary['ProfitMargin'] = (vendor_sales_summary['GrossProfit'] / vend
```

```
In [20]: vendor_sales_summary['StockTurnover'] = vendor_sales_summary['TotalSalesQuantity']
```

```
In [21]: vendor_sales_summary['SalestoPurchaseRatio'] = vendor_sales_summary['TotalSalesDoll
```

```
In [23]: vendor_sales_summary.columns
```

```
Out[23]: Index(['VendorNumber', 'VendorName', 'Brand', 'Description', 'Volume',
        'PurchasePrice', 'ActualPrice', 'TotalPurchaseQuantity',
        'TotalPurchaseDollars', 'TotalSalesQuantity', 'TotalSalesDollars',
        'TotalSalesPrice', 'TotalExciseTax', 'FreightCost', 'GrossProfit',
        'ProfitMargin', 'StockTurnover', 'SalestoPurchaseRatio'],
        dtype='object')
```

```
In [24]: cursor = conn.cursor()
```

```
In [25]: cursor.execute("""
CREATE TABLE vendor_sales_summary (
    VendorNumber INT,
    VendorName VARCHAR(100),
    Brand INT,
    Description VARCHAR(100),
    PurchasePrice DECIMAL(10,2),
    ActualPrice DECIMAL(10,2),
    Volume REAL,
```

```
TotalPurchaseQuantity INT,  
TotalPurchaseDollars DECIMAL(15,2),  
TotalSalesQuantity INT,  
TotalSalesDollars DECIMAL(15,2),  
TotalSalesPrice DECIMAL(15,2),  
TotalExciseTax DECIMAL(15,2),  
FreightCost DECIMAL(15,2),  
GrossProfit DECIMAL(15,2),  
ProfitMargin DECIMAL(15,2),  
StockTurnover DECIMAL(15,2),  
SalesToPurchaseRatio DECIMAL(15,2),  
PRIMARY KEY (VendorNumber, Brand)  
);  
""")
```

Out[25]: <sqlite3.Cursor at 0x1c894f7a4c0>

In [26]: `pd.read_sql_query("""SELECT * FROM vendor_sales_summary""", conn)`

Out[26]:

VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume	T
--------------	------------	-------	-------------	---------------	-------------	--------	---

In [27]: `vendor_sales_summary.to_sql('vendor_sales_summary',conn,if_exists='append',index=Fa`

Out[27]: 10693

In []: